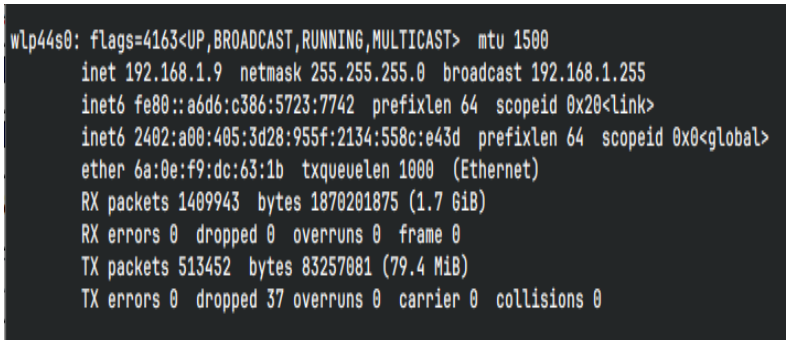
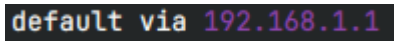
NETWORK PERFORMANCE AND PACKET ANALYSIS ASSIGNMENT

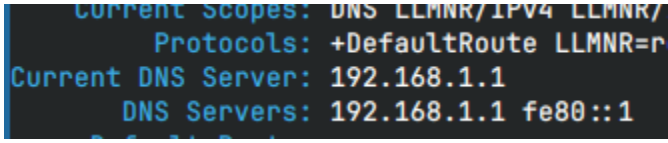
1. Run ipconfig /all in Windows Command Prompt and record your IPv4 address, subnet mask, default gateway, and DNS servers. Take a screenshot of the output.

ipconfig /all

Network detail	Your result
IPv4 address	192.168.1.9
Subnet mask	255.255.255.0
Default gateway	192.168.1.1
DNS server(s)	192.168.1.1







2. Disconnect Wi-Fi or unplug Ethernet, ping www.google.com, reconnect, and ping again. Compare the results and explain what each status means.

ping www.google.com

Expected comparison: When disconnected, ping may show 'Destination host unreachable', 'Request timed out', or a name-resolution error. After reconnecting, successful replies should display response times in milliseconds.

Network condition	Observed result / status	Meaning
Wi-Fi/Ethernet disconnected	“ping: google.com: Temporary failure in name resolution”	The computer cannot reach the destination because its network connection is unavailable.
Wi-Fi/Ethernet reconnected	“--- google.com ping statistics --- 7 packets transmitted, 7 received, 0% packet loss, time 6009ms rtt min/avg/max/mdev = 15.180/30.694/46.740/10.452 ms”	Successful replies indicate connectivity has been restored; record the reply time.

3. Monitor network usage for 5 minutes while streaming YouTube using Task Manager or Resource Monitor. Record peak network usage and the process using the most bandwidth.

Observation	Your result
Peak network usage	7.3Mbps
Process using the most bandwidth	Chrome browser
Observation duration	5 minutes

4. Capture at least 30 seconds of traffic in Wireshark while browsing Flipkart or Instagram Web. Filter HTTP or HTTPS packets and list 3 types of information visible.

Procedure

- Open Wireshark and select the active network interface.
- Start capturing, browse Flipkart or Instagram Web for at least 30 seconds, then stop the capture.
- Use the display filter http for unencrypted HTTP packets or tls to view common HTTPS/TLS traffic.
- Select a packet and inspect the packet details and packet bytes panes.

Three types of information you may see: 1. Source and destination IP addresses. 2. Transport details such as TCP source/destination ports and packet lengths. 3. For HTTPS/TLS, handshake information and metadata such as server name may be visible when included in the handshake; encrypted page content is not normally readable.

5. A friend has slow internet while gaming online. Generate a troubleshooting checklist of at least 5 actions and highlight one suggestion to try first.

- Use a wired Ethernet connection instead of Wi-Fi where possible to reduce interference and latency.
- Restart the router and modem to clear temporary connection problems.
- Pause downloads, cloud backups, and video streaming on other devices while gaming.
- Move closer to the Wi-Fi router and use the 5 GHz band (or 6 GHz if supported) to reduce interference.
- Choose the nearest available game server or region to reduce ping and latency.
- Update router firmware and computer network adapter drivers.
- Check ping and packet loss; contact the ISP if high latency or packet loss continues.
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Suggestion I would try first: Use a wired Ethernet connection. It usually provides a more stable connection with less wireless interference, which can help reduce lag and packet loss during online gaming.

Conclusion: These activities help identify network configuration, connectivity, bandwidth usage, and packet-level details. Combining command-line tools, system monitoring, and Wireshark can help diagnose common network performance problems.